Answer Key 1: Externalities - Taxation

a. To determine the socially efficient quantity, set $MSC=75$:

$50+1.5\times q=75$

$1.5\times q=25$

$q\approx 16.67$

b. To find the company's production in the absence of regulation, set $MC=75$:

$50+0.5\times q=75$

$0.5\times q=25$

$q\approx 50$

c. The Pigovian tax rate is calculated by finding the marginal external cost at the socially efficient quantity:

$MSC(16.67)-MC(16.67)=(50+1.5\times 16.67)-(50+0.5\times 16.67)=\approx 16.67$

Therefore, the Pigovian tax rate required to achieve the socially efficient quantity is approximately 16.67.

Problem 2: Positive Externalities - Subsidies

A local government is considering subsidizing a public transportation system to encourage use and reduce traffic congestion, which has positive externalities. The marginal social benefit (MSB) of the transportation system is given by $MSB=200-2q$, where q represents the number of users. The marginal private benefit (MPB), representing user demand, is $MPB=150-2q$. The marginal cost of operating the system is constant at 40.

a. Determine the socially optimal number of users (socially efficient quantity).

b. What quantity will occur without any subsidy?

c. Calculate the subsidy per user required to achieve the socially optimal quantity.

Answer Key 2: Positive Externalities - Subsidies

a. To find the socially optimal number of users, set $\boxed{MSB=40}$:

$\boxed{200-2\times q=40}$

$\boxed{2\times q=160}$

$\boxed{q=\approx 80}$

b. The quantity that occurs without any subsidy is determined by setting $\boxed{MPB=40}$:

$\boxed{150-2\times q=40}$

$\boxed{2\times q=110}$

$\boxed{q=\approx 55}$

c. The subsidy per user required to achieve the socially optimal quantity:

$\boxed{MSB(80)-MPB(80)=(200-2\times 80)-(150-2\times 80)=40-(-10)=50}$

Thus, the subsidy per user required to achieve the socially optimal quantity is 50.

Problem 3

1) Consider the market for college education in the United States. College degrees have become quite expensive in the U.S., but around 1 million people in the U.S. earn a degree each year. Suppose the yearly supply and demand functions for degrees in the U.S. are:

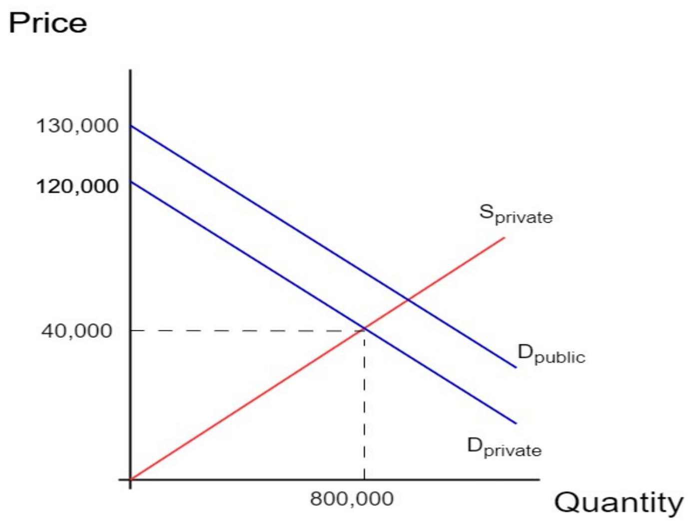
$$S(p) = 20p$$
$$D(p) = 1{,}200{,}000 - 10p$$

a. Using these supply and demand functions, find the equilibrium price and quantity in the private market for 4-year degrees.

Hopefully, students have the hang of these problems:

$$1{,}200{,}000 - 10p = 20p$$
$$1{,}200{,}000 = 30p$$
$$p* = \$40{,}000$$
$$Q^* = 20 \times 40{,}000 = 800{,}000$$

b. Suppose each person with a college degree creates an external benefit for society that can be valued at $10,000. Graph the private supply and demand curves, and label the equilibrium that you found in part a. Then, add to your graph the *social demand curve* for college degrees.



c. Label the section of your graph showing the *deadweight loss* from the externality of a college degree. Is this DWL coming from over- or underproduction?

DWL is present in the triangle bounded on the left by the quantity of 800,000, on top by the social demand line, and on bottom by the private supply line. Here, the market is underproducing the good, as buyers of college degrees do not take into account the external benefit their education produces.

d. If the government were to try to correct this market failure, would it be better to use a tax to do so, or a subsidy? What is the correct value for the tax or subsidy they should use?

With positive externalities, the government should use a subsidy to encourage consumption. The correct value for the subsidy is always exactly equal to marginal external cost: here, \$10,000.

e. When taking the external benefits of the externality into account, the social demand function for college degrees is:

$$D(p) = 1{,}300{,}000 - 10p$$

Using this new demand curve, calculate the socially optimal price and quantity for a 4-year college degree.

Again, we just set D(p) and S(p) equal to each other

$$1{,}300{,}000 - 10p = 20p$$

$$1{,}300{,}000 = 30p$$

$$p* = \$43{,}333$$

$$Q^* = 20 \times 43{,}333 = 866{,}667$$

f. If the government does not intervene in the market for college degrees, what is the dollar value of the deadweight loss in the private equilibrium?

We know the level of underproduction now (866,667-800,000), and the "base" of the externality DWL triangle is equal to the marginal external benefit. So, using our area of a triangle formula:

$$66{,}667 * 10{,}000 * .5 = \$333{,}335{,}000$$